SPECIALIST MATHEMATICS

Written examination 1

2025 Trial Examination

SOLUTIONS

Question 1 (3 marks)

$\frac{dP}{dt} = kP$

$\frac{dt}{dP} = \frac{1}{kP}$

$t = \int \left(\frac{1}{kP}\right) dP$

$t = \frac{1}{k}\log_e P + c$

1 W

$\log_e P = k(t - c)$

$P = e^{k(t-c)}$

$P = e^{kt} \times e^{-kc}$

$t = 0, m\ P = 100$

$100 = e^0 \times e^{-kc}$

$e^{-kc} = 100$

$P = 100e^{kt}$

1 W

$t = 200, P = 25$

$25 = 100e^{200k}$

$e^{200k} = 0.25$

$200k = \log_e 0.25$

$200k = \log_e\left(\frac{1}{4}\right)$

$k = \frac{2}{200}\log_e\left(\frac{1}{2}\right)$

$k = -\frac{1}{100}\log_e(2)$

$n = \frac{1}{100}$

1 A

Question 2 (7 marks)

a. (1 mark)

$z_1 = 1 + \sqrt{3}\,i$

$z_1 = 2\text{cis}\left(\frac{\pi}{3}\right)$

1 A

b. (1 mark)

$z_1 = 2\text{cis}\left(\frac{\pi}{3}\right)$

$z_2 = 2\text{cis}\left(\frac{\pi}{3} + \frac{\pi}{2}\right) = 2\text{cis}\left(\frac{5\pi}{6}\right)$

$z_3 = 2\text{cis}\left(\frac{\pi}{3} + \pi\right) = 2\text{cis}\left(\frac{4\pi}{3}\right)$

$z_4 = 2\text{cis}\left(\frac{\pi}{3} + \frac{3\pi}{2}\right) = 2\text{cis}\left(\frac{11\pi}{6}\right)$

1 A

c. (2 marks)

$z_1 = 2\text{cis}\left(\frac{\pi}{3}\right)$

$(z_1)^4 = 2^4\text{cis}\left(\frac{4\pi}{3}\right)$

1 W

$= 16\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = -8 - 8\sqrt{3}\,i$

$x = -8, y = -8\sqrt{3}$

1 A

d. (3 marks)

$$\sqrt[8]{-8-8\sqrt{3}\,i} = \left(2^4\text{cis}\left(\frac{4\pi}{3}\right)\right)^{\frac{1}{8}}$$

1 W

$$\left(2^4\text{cis}\left(\frac{4\pi}{3}\right)\right)^{\frac{1}{8}} = 2^{\frac{1}{2}}\text{cis}\left(\frac{\pi}{6}\right) = \sqrt{2}\left(\frac{\sqrt{3}}{2}+\frac{1}{2}i\right) = \frac{\sqrt{6}}{2}+\frac{\sqrt{2}}{2}\,i$$

1 W, 1 A

Question 3 (3 marks)

$n = \log_2 3$

Assume n is rational.

$n = \frac{a}{b}, a \in N, b \in N$

$2^n = 3$

$2^{\frac{a}{b}} = 3$

1 W

$\left(2^{\frac{a}{b}}\right)^b = 3^b$

$2^a = 3^b$

1 W

2^a is even since any base 2 number raised to a natural number must be even.

3^b is odd since any base 3 number raised to a natural number must be odd.

This leads to a contradiction, so the original assumption must be false.

n is not rational.

1 A

Question 4 (4 marks)

$$\int_0^{\sqrt{3}} (\tan^{-1} x)\, dx = a\pi - \log_e b$$

Use integration by parts.

$u = \tan^{-1} x\,, \dfrac{dv}{dx} = 1$

$\dfrac{du}{dx} = \dfrac{1}{1+x^2}, v = x$

1 W

$$\int_0^{\sqrt{3}} u\,\frac{dv}{dx}\,dx = uv - \int_0^{\sqrt{3}} v\,\frac{du}{dx}\,dx$$

$$= [x\tan^{-1} x]_0^{\sqrt{3}} - \int_0^{\sqrt{3}} \frac{x}{1+x^2}\,dx$$

1 W

$$= \left[x\tan^{-1} x - \frac{1}{2}log_e(1+x^2)\right]_0^{\sqrt{3}}$$

1 W

$$= \sqrt{3}\tan^{-1}\sqrt{3} - \frac{1}{2}\log_e(1+3) - 0\tan^{-1} 0 + \frac{1}{2}\log_e(1+0)$$

$$= \frac{\pi\sqrt{3}}{3} - \log_e 2$$

$a = \dfrac{\sqrt{3}}{3}, b = 2$

1 A

Question 5 (3 marks)

$4x^3y + y^2 - x - 11 = 0$

$\frac{d}{dx}(4x^3y + y^2 - x - 11) = \frac{d}{dx}(0)$

$12x^2y + 4x^3\frac{dy}{dx} + 2y\frac{dy}{dx} - 1 = 0$

1 W

$\frac{dy}{dx}(4x^3 + 2y) = 1 - 12x^2y$

$\frac{dy}{dx} = \frac{1 - 12x^2y}{4x^3 + 2y}$

1 W

Point on curve is $(1,2)$

$\frac{dy}{dx} = \frac{1 - 12(1)^2(2)}{4(1)^3 + 2(2)}$

Gradient of tangent is: $\frac{-23}{8}$

1 A

Question 6 (5 marks)

a.

$E(W) = 5 \times 6.6 + 3.2 = 36.2kg$
$Var(W) = 5 \times 0.45 \ + 0 = 2.25kg$
$sd(W) = \sqrt{2.25} = 1.5kg$

1 W, 2 A

b.

$z_1 = \frac{33.2 - 36.2}{1.5} = -2$
$z_2 = \frac{37.7 - 36.2}{1.5} = 1$

1 W

$\Pr(z < -2) \approx 0.023$
$\Pr(z > 1) \approx 0.1585$

$\Pr(-2 < z < 1) \approx 0.82$

1 A

Question 7 (6 marks)

a.

$(0,0)$

1 A

b.

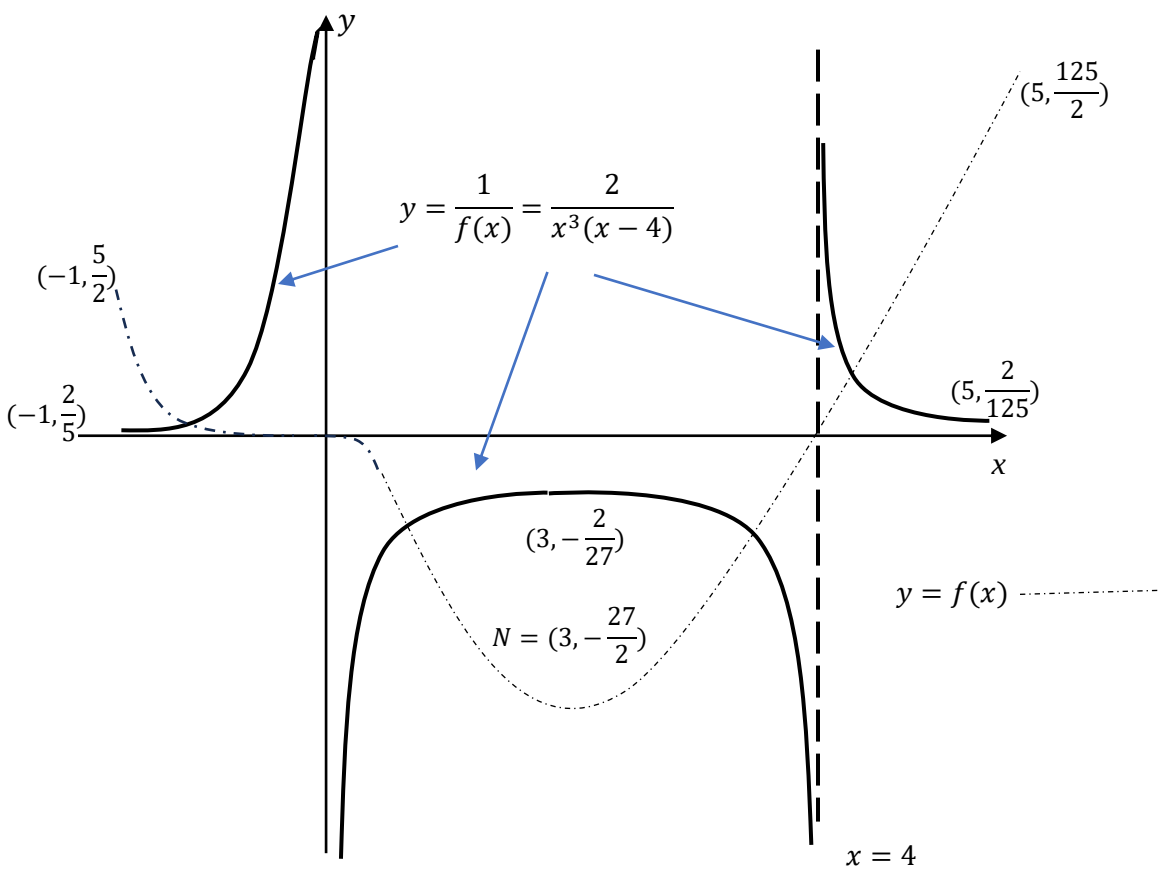
$f'(x) = 2x^3 - 6x^2$
$f''(x) = 6x^2 - 12x$
$f''(x) = 6x^2 - 12x = 0$
$6x(x-2) = 0$
$(2, -8)$ is the non-stationary POI.

1 A

Maximum negative gradient is $f'(2) = -8$.

1 A

c.



Vertical asymptotes: $x = 0, x = 4$ Horizontal asymptote: $y = 0$

1 A

Endpoints

1 A

Shape

1 A

Question 8 (3 marks)

$$V_x = \pi \int_0^{\frac{\pi}{4}} (\cos 2x)^2 \, dx$$

1 W

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{4}} (\cos 4x + 1) \, dx$$

$$= \frac{\pi}{2} \left[\frac{1}{4} \sin 4x + x \right]_0^{\frac{\pi}{4}}$$

1 W

$$= \frac{\pi}{2} \left(\frac{1}{4} \sin \pi + \frac{\pi}{4} - 0 - 0 \right)$$

$$= \frac{\pi^2}{8}$$

1 A

Question 9 (6 marks)

a.

$$\overrightarrow{PQ} = \overrightarrow{PO} + \overrightarrow{OQ}$$
$$= -2\underset{\sim}{i} + \underset{\sim}{j} + 5\underset{\sim}{k}$$
$$\overrightarrow{PR} = \overrightarrow{PO} + \overrightarrow{OR}$$
$$= -\underset{\sim}{i} - 2\underset{\sim}{j} + 0\underset{\sim}{k}$$

1 A

$$\overrightarrow{PQ} \times \overrightarrow{PR} = (0 - -10)\underset{\sim}{i} - (0 - -5)\underset{\sim}{j} + (4 - -1)\underset{\sim}{k} = 10\underset{\sim}{i} - 5\underset{\sim}{j} + 5\underset{\sim}{k}$$

1 A

b.

$$\underset{\sim}{\hat{n}} = \frac{1}{\sqrt{6}} \left(2\underset{\sim}{i} - \underset{\sim}{j} + \underset{\sim}{k} \right)$$

1 A

c.

Use any point on the plane, say P, and $\underset{\sim}{\hat{n}}$.

$d = \left|\overrightarrow{SP} \cdot \underset{\sim}{\hat{n}}\right|$

$\overrightarrow{SP} = \overrightarrow{SO} + \overrightarrow{OP} = -2\underset{\sim}{i} + 2\underset{\sim}{j} + 0\underset{\sim}{k}$

1 W

$\overrightarrow{SP} \cdot \underset{\sim}{\hat{n}} = \left(-2\underset{\sim}{i} + 2\underset{\sim}{j} + 0\underset{\sim}{k}\right).\left(\frac{1}{\sqrt{6}}\left(2\underset{\sim}{i} - \underset{\sim}{j} + \underset{\sim}{k}\right)\right)$

1 W

$= \frac{1}{\sqrt{6}}(-4 - 2 + 0) = -\frac{6}{\sqrt{6}} = -\sqrt{6}$

$d = \left|\overrightarrow{SP}\,\underset{\sim}{\hat{n}}\right| = \left|-\sqrt{6}\right| = \sqrt{6}$

1 A